

Summary of Unit-IV

TRANSPORT LAYER

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Outline

I. Introduction to Transport Layer

- *Transport Layer Services*
- *Flow control & Error Control*
 - *Simple Protocol*
 - *Stop and Wait Protocol*
 - *Go Back N (ARQ)*
 - *Selective Repeat*

II. Transport Layer Protocols

- *UDP*
- *TCP*
- *SCTP*

Summary- *I. Introduction to Transport Layer*

- Introduced the idea behind a transport-layer protocol. Discussed the general services of transport layer, such as process-to-process communication, addressing, multiplexing and de-multiplexing, error, flow, and congestion control.
- Explained the two categories of transport Layer: connectionless and connection-oriented.
- Discussed general transport-layer protocols that perform flow and error control services. Understanding these protocols helps us better understand the design of the transport-layer protocols in the Internet, such as UDP, TCP, and SCTP.

Summary- *II. Transport Layer Protocols*

- Introduced the three transport-layer protocols in the Internet and gives some information common to all of them.
- Described UDP, which is the simplest of the three protocols. UDP lacks many services we require from a transport-layer protocol, but its simplicity is very attractive to some applications
- Discussed TCP. Elaborated the services and features. Using a transition diagram, shown how TCP provides a connection-oriented service. Explained how flow and error control are accomplished in TCP. Congestion control in TCP is discussed finally.

Summary- *II. Transport Layer Protocols*

- Discusses SCTP. Explained its services and features. Shown how STCP creates an association. Explained how flow and error control are accomplished in SCTP using SACKs.

Self Assessment Questions

1. Transport layer aggregates data from different applications into a single stream before passing it to _____

- a) network layer
- b) data link layer
- c) application layer
- d) physical layer

Self Assessment Questions

2. Identify when the sender window size of Stop and Wait protocol and the 'Go back N' protocol are same with respect to the arithmetic $2m$?

Self Assessment Questions

3. Which of the following are transport layer protocols used in networking?

- a) TCP and FTP
- b) UDP and HTTP
- c) TCP and UDP
- d) HTTP and FTP

Self Assessment Questions

4. User datagram protocol is called connectionless because _____

- a) all UDP packets are treated independently by transport layer
- b) it sends data as a stream of related packets
- c) it is received in the same order as sent order
- d) it sends data very quickly

Thank You